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## ECHOPULSE\_CLEANED.pdf - 6/26/2026

EchoPulse AI: A Hybrid Local Data Architecture for Multimodal Music Recommendation Integrating WASABI, MSD, and TalkPlay Datasets

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**Abstract.** Music recommendation systems face a persistent problem: emerging artists rarely appear in suggestions. The reason is not necessarily the quality of their music. Most algorithms prioritize popularity signals over content affinity, creating feedback loops that concentrate attention on established hits. This paper presents EchoPulse AI, a locally-deployed multimodal recommender designed to address this visibility gap. The system processes lyrical, acoustic, metadata, and conversational modalities through a Triple-Encoder Transformer architecture, consolidating heterogeneous datasets (WASABI, Million Song Dataset, Lyrics 1950-2019, and TalkPlayData-2) via a five-stage local pipeline. The choice of local deployment over cloud infrastructure was driven by cost constraints and internet instability common in regions like Popayán, Colombia. Perhaps the most challenging aspect was unifying the Master Table—synthesizing information from multiple sources into a curated dataset of 5,710 tracks with complete multimodal coverage. Experimental evaluation against collaborative filtering, BERT4Rec, and SASRec baselines yielded results that exceeded initial expectations. EchoPulse AI achieves competitive accuracy ( $\text{Hit}@10 = 0.312$ ) while substantially improving Intra-List Diversity by 45.3% and, most notably, Emerging Artist Discovery by 157.9%. Core retrieval components achieve sub-250ms latency on consumer hardware. The current implementation has notable limitations: it primarily recommends English-language content, requires adaptation for Spanish-language music, and needs further development to become adaptable to arbitrary music databases. Scaling to industrial-scale catalogs and conducting online user studies remain important directions for future work.

**Keywords:** Multimodal recommendation · Transformer architecture · Local data lakehouse · Music information retrieval · Cold-start problem · Conversational AI · Generative evaluation

## 1 Introduction

Digital streaming reshaped the music business entirely. Global revenues hit USD 29.6 billion in 2024, reflecting massive growth in the streaming era [37]. But for

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independent musicians, this economic boom often feels like an illusion. The first author has spent 14 years playing drums in various bands across Colombia, and the reality on the ground is stark: new artists simply do not get recommended. Not because their music lacks quality. Because algorithms systematically favor established acts.

Most platforms still lean heavily on collaborative filtering. They just look at what people already click on. It works fine for global superstars, but it creates a brutal cold-start problem for anyone else [24, 36]. If a new track has no play history, the algorithm ignores it. This creates a vicious feedback loop. In Latin American music scenes, the situation is even worse. Local artists are not just competing with global hits; they are drowning in a sea of algorithmically-generated background music.

Content-based approaches offer a way out. By looking at the actual lyrics, acoustic properties, and metadata of a song, it becomes possible to recommend based on what the music actually sounds like [35]. Comprehensive treatments of recommender systems theory [34] provide the foundation upon which modern Transformer-based approaches are built. The seminal work "Attention Is All You Need" by Vaswani et al. [1] particularly inspired this project. Its ability to map long-range dependencies makes Transformers ideal for capturing the complex structure of musical content [40].

This paper introduces EchoPulse AI. It was built as a locally-deployed multi-modal recommender within the IMS Research Group at Fundación Universitaria de Popayán. Why local? The answer is simple. The research group had a limited budget for cloud services, and internet connectivity in the region is notoriously unstable. Instead of seeing these constraints as a weakness, the team used them to force a design that prioritizes efficiency, reproducibility, and zero cloud dependency.

Three main contributions emerge from this work. First, a hybrid local data architecture is proposed. It follows a five-stage paradigm to merge messy, heterogeneous music datasets without needing AWS or GCP. Second, a Triple-Encoder Transformer is designed. It processes lyrics, audio, and conversational data through specialized modules, backed by strict mathematical formulations. Third, an end-to-end evaluation is provided. The results prove that it is possible to drastically improve diversity and surface emerging artists without tanking accuracy metrics.

Building the Master Table was easily the hardest part of this project. The team had to reconcile completely different ID systems, handle massive amounts of missing data, and force four distinct datasets into a single, clean collection of 5,710 tracks. That data engineering struggle directly shaped the architecture described in the following sections.

The rest of the paper walks through the approach step-by-step. Section 2 covers the background on Transformers and local data architectures. Section 3 dives into the five-stage pipeline and the model math. The hardware setup and metrics are detailed in Section 4, followed by an honest look at the limitations

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in Section 5. Finally, Section 6 wraps up the findings and what they mean for the future of equitable music recommendation.

## 2 Related Work

### 2.1 Transformer-Based Recommendation

The introduction of the Transformer architecture by Vaswani et al. [1] revolutionized sequence modeling through the self-attention mechanism:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax} \left( \frac{\mathbf{Q} \mathbf{K}^{\text{top}}}{\sqrt{d_k}} \right) \mathbf{V}$$

where  $\mathbf{Q}$ ,  $\mathbf{K}$ , and  $\mathbf{V}$  represent query, key, and value matrices respectively, and  $d_k$  denotes the dimensionality of key vectors. This formulation enables models to capture long-range dependencies in sequential data, making it particularly suitable for modeling musical structure and user listening patterns. BERT4Rec [4] introduced bidirectional encoding for sequential recommendation, improving accuracy but inheriting popularity bias from training data. Similarly, SASRec [3] leveraged self-attention for sequential recommendation, establishing strong baselines for comparison.

### 2.2 Multimodal Music Recommendation

Multimodal approaches that integrate multiple data sources have shown promise for improving recommendation quality and diversity [40]. Foundational work on musical genre classification [28] established the feature extraction pipelines that modern multimodal systems build upon. Research has explored fusion techniques for combining audio features with textual metadata [12], while cross-modal attention mechanisms have been investigated for music understanding [18, 19]. The WASABI dataset [6] and Million Song Dataset [5] provide foundational resources for multimodal music research, enabling large-scale analysis of lyrical, acoustic, and metadata features. Recent work on music emotion recognition leverages the circumplex model of affect [42, 30] to map acoustic and lyrical features to emotional dimensions.

### 2.3 Local Data Architecture for ML Systems

The Lakehouse paradigm has emerged as a modern approach to data management, combining the flexibility and scalability of data lakes with the transactional capabilities of data warehouses [15]. While technologies such as Delta Lake, Apache Iceberg, and Apache Hudi enable ACID transactions over object storage, the implementation in this work adapts these principles to a local environment using parquet-based storage with manual versioning, schema validation, and snapshot isolation—providing reproducibility and traceability without cloud infrastructure dependencies [45].

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#### EchoPulse AI Multimodal Music Recommender System Architecture

Fig. 1. EchoPulse AI Local Data Architecture: End-to-end pipeline from heterogeneous sources to local deployment with evaluation feedback loop. The five stages—Data Sources, Data Lake, Data Processing, Data Modeling, and Data Product—ensure reproducibility and traceability without cloud dependencies. Source: Own elaboration.

## 2.4 Conversational Recommendation Systems

Recent work such as TalkPlay [8] employs large language models for music recommendation through natural language interfaces. However, these approaches often lack structured multimodal integration and rigorous evaluation of generative components. The present work extends this direction by incorporating a dedicated conversational encoder trained on the TalkPlayData-2 dataset (~16,000 structured dialogues) and proposing evaluation metrics specific to intention-aware recommendation [26, 27]. Retrieval-augmented generation frameworks [16, 17] provide methodological foundations for integrating external knowledge sources with generative models.

## 3 Methodology

### 3.1 Data Architecture Overview: Five-Stage Local Pipeline

The EchoPulse AI system implements a five-stage data pipeline following established MLOps principles, adapted for local deployment. Figure 1 illustrates the end-to-end architecture from heterogeneous data sources to local deployment with an evaluation feedback loop.

### 3.2 Data Sources and Integration

The system integrates four primary data sources, each contributing distinct modalities:

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WASABI Dataset [6]: Provides lyrical content and cultural metadata for over 2 million commercially released songs, with temporal coverage spanning 1922 to 2022.

Million Song Dataset (MSD) [5]: Offers 26 continuous acoustic features including tempo ( $\tau$ ), energy (E), valence (V), danceability (D), and loudness

● Sentences that are likely AI-generated.

## FAQs

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### What is GPTZero?

GPTZero is the leading AI detector for checking whether a document was written by a large language model such as ChatGPT. GPTZero detects AI on sentence, paragraph, and document level. Our model was trained on a large, diverse corpus of human-written and AI-generated text with support for English, Spanish, French, German, and other languages. To date, GPTZero has served over 17 million users around the world, and works with over 100 organizations in education, hiring, publishing, legal, and more.

### When should I use GPTZero?

Our users have seen the use of AI-generated text proliferate into education, certification, hiring and recruitment, social writing platforms, disinformation, and beyond. We've created GPTZero as a tool to highlight the possible use of AI in writing text. In particular, we focus on classifying AI use in prose. Overall, our classifier is intended to be used to flag situations in which a conversation can be started (for example, between educators and students) to drive further inquiry and spread awareness of the risks of using AI in written work.

### Does GPTZero only detect ChatGPT outputs?

No, GPTZero works robustly across a range of AI language models, including but not limited to ChatGPT, GPT-5, GPT-4, GPT-3, Gemini, Claude, and AI services based on those models.

### What are the limitations of the classifier?

The nature of AI-generated content is changing constantly. As such, these results should not be used to punish students. We recommend educators to use our behind-the-scenes [Writing Reports](#) as part of a holistic assessment of student work. There always exist edge cases with both instances where AI is classified as human, and human is classified as AI. Instead, we recommend educators take approaches that give students the opportunity to demonstrate their understanding in a controlled environment and craft assignments that cannot be solved with AI. Our classifier is not trained to identify AI-generated text after it has been heavily modified after generation (although we estimate this is a minority of the uses for AI-generation at the moment). Currently, our classifier can sometimes flag other machine-generated or highly procedural text as AI-generated, and as such, should be used on more descriptive portions of text.

### I'm an educator who has found AI-generated text by my students. What do I do?

Firstly, at GPTZero, we don't believe that any AI detector is perfect. There always exist edge cases with both instances where AI is classified as human, and human is classified as AI. Nonetheless, we recommend that educators can do the following when they get a positive detection: Ask students to demonstrate their understanding in a controlled environment, whether that is through an in-person assessment, or through an editor that can track their edit history (for instance, using our [Writing Reports](#) through Google Docs). Check out our list of [several recommendations](#) on types of assignments that are difficult to solve with AI.

Ask the student if they can produce artifacts of their writing process, whether it is drafts, revision histories, or brainstorming notes. For example, if the editor they used to write the text has an edit history (such as Google Docs), and it was typed out with several edits over a reasonable period of time, it is likely the student work is authentic. You can use GPTZero's Writing Reports to replay the student's writing process, and view signals that indicate the authenticity of the work.

See if there is a history of AI-generated text in the student's work. We recommend looking for a long-term pattern of AI use, as opposed to a single instance, in order to determine whether the student is using AI.